# Lab *04 – Searching and Filtering (ethically!)*

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## INTRODUCTION

In this lab, I am going to be scouring the web, looking for anything related to, uh. Me. I am doing an ethical search on an entity which I have permission (again, me, hi there) using a few different methods to discover just how visible my information is, including my email, phone number, passwords, and more.

## PROCESS

### Breakpoint 2: Find Your Details Online

I, first, used both the haveibeenpwned website to check the visibility of my emails and my phone number. I was over the moon when I saw my official email and phone number were not visible in any data breach, BUT my old email that I have most of my game passwords and other minor things on had sadly been pwned. It appeared in a data breach at some point, so I went back and changed the password on it and added two-factor authentication to the account.

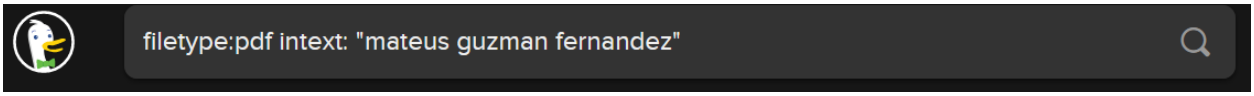
Our token family password, however, was not so lucky. I looked it up on haveibeenpwned (for passwords), and discovered that it had been seen 16 times before in data breaches, so we have a bit of a situation to take care of here. My other passwords, however, for a few of the things I used the most now were all secure and were not detected in data breaches.

Then, I checked out my IP addresses on whatismyipaddress. My IPv4 address was interestingly the same on my virtual machine and actual device. I would be able to explain why last year, but I honestly forgot. I guess since the VM and home device run on the same hardware, the IP address does not differ, whereas my desktop at home would have another IP. My VM did not have an IPv6 address, but I am not sure why, either.

Lastly, I looked at my MAC address on whatsmyip. I thought there would be more information displayed on it, but what I found was still interesting. Intel must have a standard MAC address shell because the site only displayed 68:3e:26 and showed Intel Corporate as the vendor's name as the only bit of information.

## Breakpoint 3: Google Hacking

These are the dorks I used:

1. 


filetype indicates the type of file you want to filter for, and intext filters for the specific text you input. In my case, I told my search engine to search for my name in the texts of PDF files online. Thankfully, nothing came up.

Sources:

https://kit.exposingtheinvisible.org/en/how/google-dorking.html

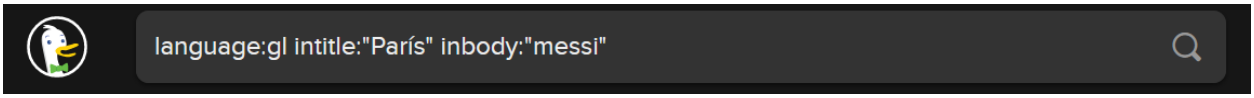
2.


language tells the search engine to search in a specific language based off the inputted language code (you can find these here). I chose Galician because of my interest in the region, and its code was gl. intext tells it to search for anything in its index that has my input "santiago de compostela" in its text contents.

Sources:

https://www.loc.gov/standards/iso639-2/php/code_list.php

https://kit.exposingtheinvisible.org/en/how/google-dorking.html

3. 


language tells the search engine to search in a specific language based off the inputted language code (you can find these here). intitle searches for your input in the title of the page, and typed París with the accented "I" since that is how it is written in Galician. Lastly, I chose inbody:"messi" to search for anything in the body text of media relating to Messi. I am a huge football fan and I thought it would be interesting to see what Galician media had to say about Lionel Messi's transfer to Paris Saint-Germain.

Sources:

https://www.compass-security.com/fileadmin/Research/White_Papers/2017-01_osint_cheat_sheet.pdf

https://kit.exposingtheinvisible.org/en/how/google-dorking.html

## Breakpoint 4: Grepping Your Slack Workplace

*I had trouble accessing my Slack account, so I am using my professor's name to do this section.

First, I used **grep -R "rita.mitra"** to search all the folders recursively, which checks all files, folders, and subdirectories in the chosen directory for your input. The quotes are important since they denote your input as a string. This showed me every instance my professor sent a message in our Slack workspace as the output, which I represented below in a considerably shorter snip.

```
mgfubuntu@MateusGuzmanFernandez-VirtualBox:~/Downloads/slack$ grep -R "rita.mitra"
module-06/2021-10-29.json:            "name": "rita.mitra",
module-06/2021-10-29.json:            "name": "rita.mitra",
module-06/2021-10-24.json:            "name": "rita.mitra",
module-06/2021-10-28.json:            "name": "rita.mitra",
module-06/2021-10-30.json:            "name": "rita.mitra",
```

Then, I used **grep -R '"name": "rita.mitra"' | wc -l** to discover how many times her name appeared in our Slack. The pipe **|** adds a command to your line here, and I added **wc -l (wordcount -l)** along with the original **grep** command to also find out how many times it came up, print the names of the files and the number of lines in those files it appeared in **(-l)**. It turns out that her name appeared 667 times according to my restrictions.

```
mgfubuntu@MateusGuzmanFernandez-VirtualBox:~/Downloads/slack$ grep -R '"name": "rita.mitra"' | wc -l
667
```

Next, I printed all the HTTPS links in our workspace by using **grep -R "<https:"**. **<** denotes an external link in Slack.

```
mgfubuntu@MateusGuzmanFernandez-VirtualBox:~/Downloads/slack$ grep -R '<https:'
module-06/2021-10-29.json:            "text": "Guido van Rossum, Python's inventor, pointed to an explanation of indentation in Python on <https:\/\/twitter.com\/gvanrossum\/status\/1249549091584892928?lang=
en|this twitter feed>. Thanks for trying out the code ad bringing up this important and interesting issue! Simply typing in code and running it leads to good insights about how code works :slightly_smilin
g_face: . (Also, take a look at the <https:\/\/archive.org\/web\/|Wayback Machine> cited here. Pretty interesting archival project!).",
module-07/2021-11-03.json:            "text": "Hi all: USC is inviting UTSA students, faculty, and staff to join them for a free, Election Cybersecurity Workshop for Texas and adjacent states on Thursday, No
vember 4, from 1:30 to 3:30 pm Central Time. I'm posting the announcement on this channel because the topic is misinformation\/disinformation, which is relevant here and in Module 08.\n\n\"As we have see
n in the pipeline and JBS meat attacks, cybersecurity and ransomware threats are evolving, and disinformation tools are changing, so we will have the latest from experts in Washington DC and Los Angeles.
 The program will also include free security resources available from government and industry.\"\n \nDetails and RSVP are at *<https:\/\/uscninthregional.eventbrite.com\/?aff=universities>*",
module-07/2021-10-11.json:            "text": "60 Minutes has an <https:\/\/www.cbsnews.com\/news\/deepfake-artificial-intelligence-60-minutes-2021-10-10\/|interesting segment on Deep Fakes> last night that
could be relevant here or in the Ethics module. It's quite sobering to think that our identity and actions can be simulated on our behalf!",
```

After that, I saved all the text in Slack using **grep -R '"text":'**, which gave me so much output that I could not scroll to the top.

```
lab-02/2021-10-19.json:            "text": "im really confused on what to do on phase 12 at this point",
lab-02/2021-10-19.json:                                    "text": "im really confused on what to do on phase 12 at this point"
lab-02/2021-10-19.json:            "text": "<@U02BZ675RQT> You are looking to find a way to create a sub branch on your local (Your Linux\/Desktop), and then send it back to your remote (GitHub).  This step
requires you to do some Sherlock Holmes work  (research and investigate to find out how).  There were some pointers on how to do that search a bit back from <@U02BASNJN8N>.  I always find that short litt
le how to videos or pictures help me, but not everyone is the same.  There are some guides I found that were helpful too.  I hope that helps, if not, please be a little more specific on what you are conf
used on, so we can help you better :)",
lab-02/2021-10-19.json:                                    "text": " You are looking to find a way to create a sub branch on your local (Your Linux\/Desktop), and then send it back to your remote (GitHub).
This step requires you to do some Sherlock Holmes work  (research and investigate to find out how).  There were some pointers on how to do that search a bit back from "
lab-02/2021-10-19.json:                                    "text": ".  I always find that short little how to videos or pictures help me, but not everyone is the same.  There are some guides I found that wer
e helpful too.  I hope that helps, if not, please be a little more specific on what you are confused on, so we can help you better :)"
```

I redirected the saved text from the previous command to another folder using **grep -R '"text":' >> horatioslack.txt** for future formatting.

```
mgfubuntu@MateusGuzmanFernandez-VirtualBox:~/Downloads/slack$ grep -R '"text":' >> horatioslack.txt
grep: input file 'horatioslack.txt' is also the output
```

Onto **regexes**, or regular expressions. These are powerful search shortcuts we can add to search commands to quickly match strings and characters. For example, if we want to search for a specific word, and it is referred to in both letter cases, you do not have to do two individual searches for the same word with different capitalizations. You can use a **regex** , **[ ]**, to put in both the cases of a letter to search for in your input's directory. In my example, I used **grep -R "[gG]it" lab-02** to search for any instance where the word **git** or **Git** was used in the lab-02 folder's chat logs.

```
lab-02/2021-10-12.json:        "text": "On phase 1 when I made my github education thingy it says thanks for submitting, be sure to check your email. Am I supposed to wait for an email from them or can I
 go on with the lab?",
lab-02/2021-10-12.json:                                "text": "On phase 1 when I made my github education thingy it says thanks for submitting, be sure to check your email. Am I supposed to wait for an
email from them or can I go on with the lab?"
lab-02/2021-10-12.json:        "text": "Have you created an account on GitHub, and just not the verification? If so, you can go ahead, I believe. I'm actually curious to see how it's working right now, s
o let us know.",
lab-02/2021-10-12.json:                                "text": "Have you created an account on GitHub, and just not the verification? If so, you can go ahead, I believe. I'm actually curious to see how i
```

Now, I changed the command to only show **GitHub** and its variations, not just **git** or **Git**, by using **grep -R "[gG]it[hH]ub" lab-02**.

```
mgfubuntu@MateusGuzmanFernandez-VirtualBox:~/Downloads/slack$ grep -R "[gG]it[hH]ub" lab-02
lab-02/2021-10-12.json:        "text": "For phase 8 I am not sure why I got this error after I entered my password. fatal: Authentication failed for <https:\/\/github.com\/aditi-shinde\/IS1003-firstrepo.
git>",
lab-02/2021-10-12.json:                                "text": "https:\/\/github.com\/aditi-shinde\/IS1003-firstrepo.git",
lab-02/2021-10-12.json:        "text": "Hm I tried again but I keep having the same problem. I tried to solve it looking at github docs but no luck. I'm thinking to completely start over.",
lab-02/2021-10-12.json:                                "text": "Hm I tried again but I keep having the same problem. I tried to solve it looking at github docs but no luck. I'm thinking to completely sta
rt over."
lab-02/2021-10-12.json:        "text": "Hi Aditi: It may be easier and quicker to start over with a new directory\/folder\/repo. I also looked at github docs and see that you might could edit your ssh co
nfiguration file, but this is a more arduous and possibly not successful task. I myself have redone new repos many times :slightly_smiling_face: ",
```

There are other **regexes** to use with these commands that can broaden your search possibilities. One such type are **wildcards,** which are a data type that means any character put where the wildcard is in your command is valid. In the command below, I used **grep -R "[gG]it[hH]*u*b" lab-02** to search for any combination of the words **git** or **GitHub** in any letter case. The ***** wildcard can replace any combination of characters and numbers in a string, and I used it over the **.** wildcard because I needed to find two separate cases of **h** for my command. **.** only replaces one character, so I had to use ***** to find both cases of **h.**

```
mgfubuntu@MateusGuzmanFernandez-VirtualBox:~/Downloads/slack$ grep -R "[gG]it[hH]*u*b*" lab-02
lab-02/2021-10-12.json:        "text": "For phase 8 I am not sure why I got this error after I entered my password. fatal: Authentication failed for <https:\/\/github.com\/aditi-shinde\/IS1003-firstrepo.
git>",
lab-02/2021-10-12.json:                                "text": "https:\/\/github.com\/aditi-shinde\/IS1003-firstrepo.git",
lab-02/2021-10-12.json:        "text": "Hm I tried again but I keep having the same problem. I tried to solve it looking at github docs but no luck. I'm thinking to completely start over.",
lab-02/2021-10-12.json:                                "text": "Hm I tried again but I keep having the same problem. I tried to solve it looking at github docs but no luck. I'm thinking to completely sta
rt over."
```

Finally, I went and created my own command to search for common question formats to see what questions people had about **lab-03**. I used **cd** to get to **lab-03**'s folder*, then I used my command **grep -R "[wW]hat\|[wW]hen" 2021-11-01.json** recursively searches the file **2021-11-01.json** for the words **what** and **when** with **w** being in any letter case. I figured out that you can add multiple inputs to search for with **grep** using **\|**. **Grep** does not use the pipe as the **or** operator, so you need to add a backslash in front of it to remove its special status, which enables me to add another search term to my command. I added **[wW]hen** to my command to also search for anything in the file with the word **when** with the **w** being in any case.

*You do not have to be in a specific folder to use **grep** to search for something within it. If you are in the folder you just have to type less to achieve your goal, so it becomes easier. If you are not in the folder, you have to specify the path of the file you are **grep**ping at the end of your command. If I was in **lab-02**, I would have had to type **grep -R "[wW]hat\|[wW]hen" ~/Downloads/slack/lab-03/2021-11-01.json** to do the same effect.

```
mgfubuntu@MateusGuzmanFernandez-VirtualBox:~/Downloads/slack/lab-03$ grep -R  '[wW]hat\|[wW]hen' 2021-11-01.json
        "text": "I'm having a little trouble with phase 7, I've looked over it a couple times and can't figure out what I'm doing wrong. I'm not getting the right output with lab_time, all I'm getting is
 "0:12" any insight on this would be greatly appreciated ",
                                "text": "I'm having a little trouble with phase 7, I've looked over it a couple times and can't figure out what I'm doing wrong. I'm not getting the right output with lab_
time, all I'm getting is "0:12" any insight on this would be greatly appreciated "
        "text": "Can you post a screenshot of what your script says in the nano editor?",
                                "text": "Can you post a screenshot of what your script says in the nano editor?"
        "text": "In Phase 3 what is it referring to by saying \"your subsequent about-me.sh code\"",
                                "text": "In Phase 3 what is it referring to by saying \"your subsequent about-me.sh code\""
        "text": "<@U02C1L6035Y> This refers to what your code looks like after you performs the commands in this section.",
                                "text": " This refers to what your code looks like after you performs the commands in this section."
        "text": "In phase 4, are we editing the nano code to what the instructions say in about-me or about-me-v2?",
                                "text": "In phase 4, are we editing the nano code to what the instructions say in about-me or about-me-v2?"
        "text": "This relates to the question in Breakpoint #3 when you execute the script after appending the string (with the Bash shell variable) into the script. Why does it happen (see Ben's hint)?
Also, note that this issue is fixed in the next Phase by placing the entire command in the function.",
                                "text": "This relates to the question in Breakpoint #3 when you execute the script after appending the string (with the Bash shell variable) into the script. Why does it h
appen (see Ben's hint)? Also, note that this issue is fixed in the next Phase by placing the entire command in the function."
```

*Here is the command used in the sidenote

```
mgfubuntu@MateusGuzmanFernandez-VirtualBox:~/Downloads/slack/lab-02$ grep -R  '[wW]hat\|[wW]hen' ~/Downloads/slack/lab-03/2021-11-01.json
        "text": "I'm having a little trouble with phase 7, I've looked over it a couple times and can't figure out what I'm doing wrong. I'm not getting the right output with lab_time, all I'm getting is
 "0:12" any insight on this would be greatly appreciated ",
                                "text": "I'm having a little trouble with phase 7, I've looked over it a couple times and can't figure out what I'm doing wrong. I'm not getting the right output with lab_
time, all I'm getting is "0:12" any insight on this would be greatly appreciated "
        "text": "Can you post a screenshot of what your script says in the nano editor?",
                                "text": "Can you post a screenshot of what your script says in the nano editor?"
        "text": "In Phase 3 what is it referring to by saying \"your subsequent about-me.sh code\"",
                                "text": "In Phase 3 what is it referring to by saying \"your subsequent about-me.sh code\""
        "text": "<@U02C1L6035Y> This refers to what your code looks like after you performs the commands in this section.",
                                "text": " This refers to what your code looks like after you performs the commands in this section."
        "text": "In phase 4, are we editing the nano code to what the instructions say in about-me or about-me-v2?",
                                "text": "In phase 4, are we editing the nano code to what the instructions say in about-me or about-me-v2?"
        "text": "This relates to the question in Breakpoint #3 when you execute the script after appending the string (with the Bash shell variable) into the script. Why does it happen (see Ben's hint)?
Also, note that this issue is fixed in the next Phase by placing the entire command in the function.",
                                "text": "This relates to the question in Breakpoint #3 when you execute the script after appending the string (with the Bash shell variable) into the script. Why does it h
appen (see Ben's hint)? Also, note that this issue is fixed in the next Phase by placing the entire command in the function."
```

## CONCLUSION

Please provide a conclusion based on these prompts:

- What are the goals of the techniques and tools used in this lab? How are they relevant to your degree and your current or future career?
- How effective are these techniques and tools, in your opinion? What other aspects would you like to explore?
- What is your opinion of this lab in terms of difficulty and relevance?

The goals of what we used in this lab were to give us an introduction to smart searching, using the command line interface to filter for specific information in files, folders, and directories in Linux as well as ethically discovering vulnerable information that is readily available on the Internet. This is going to be extremely relevant later because I can already see how much time it will save and how we can use it to defend ourselves on the Internet to better maintain our privacy or anonymity. I think these tools are already highly effective, even though we have just scratched the surface with grepping and the wildcards we used to gather information on our workspace, but as we delve further into more wildcards

and other topics within smart searching outside of Linux, we will become extremely proficient at quickly gathering important information, an invaluable skill for our careers, especially in times of crisis in the organization like ongoing attacks. I think this lab is easy, and I do like that we were able to explore the different **grep** variations and talk a little about wildcards. It is a good, relevant, and well-paced introduction to important concepts that we will absolutely need in the future, and my favorite part was doing research to create my own **grep** command to find information.

## References

**Internet Resources**


Yet Another Firefox Hardening Guide (Xiao, 29 March 2021)

DuckDuckGo's home page

Haveibeenpwned website

Haveibeenpwned for passwords

Whatismyipaddress website

Whatsmyip (using this for my MAC address)

Google Hacking Database (GHDB)

Search Smarter by Dorking (Sobliye, Exposing the Invisible | The Kit, n.a.)

OSINT Cheat Sheet (Compass, 2017)

Google Cheat Sheet (SANS, 2006)

DuckDuckGo search operators

What is JSON? (w3schools)

Linux/Mac Terminal Tutorial: The Grep Command - Search Files and Directories for Patterns of Text (Corey Schafer, 07 March 2018)

Ryan's Tutorial: Linux Tutorial: Cheat Sheet (Ryan Chadwick, 2021)

wc command in Linux with examples (GeeksForGeeks, 19 February 2021)

How to Grep for Multiple Strings, Patterns or Words (phoenixNAP, 5 May 2020)


**Collaboration**

I did not collaborate with anyone on this lab.